

NEW AZAINDOLE DERIVATIVES AS ANALGESIC AGENT

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ABSTRACT:

The indole class of analgesics continues to be exposed to the pharmaceutical community. The synthesis and analgesic properties of these agents have been the subject of ongoing investigations. During the course of present research, effort were made to evaluate the possible analgesic activity of various quaternary ammonium salts of 7-azaindole to see whether one more nitrogen in the indole nucleus also responsible for analgesic activity.

Keywords: 7-Azaindole, Phenacyl halide Analgesia

INTRODUCTION

The analgesic drugs presently used in therapeutics are either narcotics or drugs with more limited efficacy. Both group present well known side effects and toxic effects. The lack of potent analgesic drug with severe adverse reaction prompted the scientists to struggle in this field.

A number of indole derivatives have been proved to possess potent analgesic antipyretic and anti-inflammatory activity (Shen et al., 1963 and Dumex A. 1964). Great contributions have been made by Japanese workers in order to explore the analgesic activity of various indole derivatives (Michio et al., 1968; Hisao et al., 1970 and Tsutomu et al., 1986).

Little work on analgesic, antipyretic and anti-inflammatory activity of 7-azaindole and its derivatives is recorded in the literature, however, certain derivatives of azaindole were found to possess anti-inflammatory and analgesic activity (Ting et al., 1990; Bisagni et al., 1973; Saify Z.S. 1982). A patent (Merck & Co. 1966) describing the synthesis of aza analogues of the drug indomethacin claims the 7-azaindole derivatives to have anti-inflammatory analgesic and antipyretic properties.

In our laboratory we have synthesized a series of substituted phenacyl derivatives of 7-azaindole and to study the in vivo and in vitro biological properties of these newly synthesized derivatives (Distasi et al. 1988). In this communication we have reported the analgesic screening of these new compounds in comparison with morphine HCl and parent compound. The goal of this study is to explore the potential of 7-azaindole as a biological active compound.

MATERIALS AND METHODS

General

An equimolar quantities of 7-azaindole and the corresponding substituted phenacyl halides were dissolved in acetone in separate conical flasks and mixed together in a round bottom flask. The contents were then kept on stirring for 4 hrs. the mixture was set aside a room temperature till a solid product obtained. The completion of the reaction was monitored by TLC (CHCl₃-MeOH; varying proportions) system. The resulting crude product was filtered and washed with acetone to remove the unreacted starting materials. The structural elucidation of compounds were carried out by ¹H-NMR, EIMS, UV and IR.

Pharmacology

Analgesic activity of the compounds was tested as antinociceptive effect against thermal stimuli (Tail Flick Test).

White Albino mice of either sex (locally bred) weighing between 20-25 gm were used. The animals were maintained in a group of five under standard colony conditions i.e., 12 hrs. light and 12 hrs. dark temperature $21 \pm 2^{\circ}\text{C}$. fed with balance diet and water.

Compounds were dissolved in water and injected to the test animals intramuscularly at doses of 50, 75 and 100 mg/kg body weight. Morphine HCl (10 mg/kg) was used as a standard drug and distilled water used in case of control group which were always run parallel to the compound treated group.

The method used was a modification of Distasi et. al., 1988. Mice were held in position in a suitable restrainer with the tail extending out. The one third of the tail was immersed in water bath thermostatically maintained at 51°C . The cut off time for immersion was 180 seconds to avoid the injury of tissues of tail. The reaction time was evaluated at +30, +60 and +90 minutes after the administration of compound. The mean increase in latency after drug administration or analgesia TFLD $\pm$ S.E.M. were calculated. Statistical analysis were performed using student's t-test and values were considered significant if $P < 0.05$.

RESULT

Analgesic activity of all derivatives (II-XIV) alongwith their parent compound (I) are presented in Table 1-4. Results are indicated in term of analgesia "TFLD" (Tail flick latency difference) or mean increase in latency after the drug administration $\pm$ S.E.M. in seconds. The TFLD values of all the tested compounds were compared with

water injected control mice and morphine HCl injected mice at various doses.

All the compounds possess varying degree of analgesic activity. Most of the derivatives exhibited promising activity in comparison with control, but these derivatives are less effective as that of morphine HCl.

DISCUSSION

Among all derivatives tested, 3',4'-dihydroxy derivative (IX) was proved the most active compound (Table-1). It showed the peak activity at a dose of 75 mg/kg after 60 minutes administration (4.52 ± 0.12). This compound (IX) displayed ten fold less activity then morphine HCl. Compound VIII with a bulky phenyl substituent at C-4' position was next to compound IX. It showed maximum TFLD (2.34 ± 0.15) at dose of 100 mg/kg after 90 minutes administration.

All four methoxy containing compounds (III, X-XI and XIII) found to have analgesic activity (Table-2). Among these, *para* methoxy derivative (III) was the most active at a dose of 75 mg/kg and showed maximum TFLD value (1.54 ± 0.07) after 60 minutes administration.

The halogen containing compounds (IV-VI and XII) also showed significant activity (Table-4) however, compound V and VI having mono bromo and mono fluoro substituent at *para* position were found more active. Rest of three compounds (II, VII and XIV) in this series did not show any appreciable activity at the doses used (Table-3).

The diversities in the results among the derivatives of this series can be attributed due to the presence of different substituent at different position in the phenyl part of the molecule which may affect the activity of compounds to the analgesic receptor binding.

[II]	$R^3 = CH_3$	$R^1 = R^2 = R^4 = H$	$X = Br$
[III]	$R^3 = CH_3$	$R^1 = R^2 = R^4 = H$	$X = Br$
[IV]	$R^3 = Cl$	$R^1 = R^2 = R^4 = H$	$X = Br$
[V]	$R^3 = Br$	$R^1 = R^2 = R^4 = H$	$X = Br$
[VI]	$R^3 = F$	$R^1 = R^2 = R^4 = H$	$X = Cl$
[VII]	$R^3 = NO_2$	$R^1 = R^2 = R^4 = H$	$X = Br$
[VIII]	$R^3 = Ph$	$R^1 = R^2 = R^4 = H$	$X = Br$
[IX]	$R^2 = R^3 = OH$	$R^1 = R^4 = H$	$X = Cl$
[X]	$R^1 = R^3 = OCH_3$	$R^2 = R^4 = H$	$X = Br$
[XI]	$R^1 = R^4 = OCH_3$	$R^2 = R^3 = H$	$X = Br$
[XII]	$R^1 = R^3 = F$	$R^2 = R^4 = H$	$X = Cl$
[XIII]	$R^2 = OCH_3$	$R^1 = R^3 = R^4 = H$	$X = Br$
[XIV]	$R^1 = NO_2$	$R^2 = R^3 = R^4 = H$	$X = Br$

Table 1
Analgesic Effects of 7-Azaindole (I) and its Derivatives (VIII and IX)
in Mouse Tail Flick Method

Treatment	Dose/kg	Analgesia TFLD (mean increase in Latency after drug admn. $\pm$ SEM(s))		
		30 min	60 min	90 min
Water for Injection	0.3 ml	0.19 ± 0.04	0.15 ± 0.07	0.28 ± 0.08
I	50 mg	$0.14^* \pm 0.08$	$0.24^* \pm 0.07$	0.38 ± 0.05
	75 mg	$0.22^* \pm 0.08$	0.44 ± 0.01	$0.30^* \pm 0.07$
	100 mg	0.52 ± 0.06	$0.24^* \pm 0.05$	$0.14^* \pm 0.08$
VIII	50 mg	$0.04^* \pm 0.08$	1.10 ± 0.08	0.62 ± 0.13
	75 mg	0.82 ± 0.05	1.91 ± 0.10	1.46 ± 0.12
	100 mg	0.96 ± 0.11	2.12 ± 0.11	2.34 ± 0.15
IX	50 mg	1.68 ± 0.06	2.14 ± 0.07	1.86 ± 0.07
	75 mg	1.88 ± 0.05	4.52 ± 0.12	3.72 ± 0.16
	100 mg	1.62 ± 0.05	2.28 ± 0.08	1.96 ± 0.06
Morphine HCl	10 mg	44.4 ± 0.526	48.3 ± 0.48	55.2 ± 0.44

N = 5 P < 0.05 (All the observations are significant except *Insignificant)

observations)

Table 2
Analgesic Effects of 7-Azaindole Derivatives (III, X-XI and XIII)
in Mouse Tail Flick Method

Treatment	Dose/kg	Analgesia TFLD (mean increase in Latency after drug admn. $\pm$ SEM(s))		
		30 min	60 min	90 min
Water for Injection	0.3 ml	0.36 ± 0.18	0.15 ± 0.12	0.19 ± 0.09
III	50 mg	0.56 ± 0.08	0.95 ± 0.11	0.83 ± 0.07
	75 mg	1.12 ± 0.06	1.54 ± 0.07	1.09 ± 0.05
	100 mg	0.96 ± 0.05	1.15 ± 0.05	0.65 ± 0.13
X	50 mg	0.64 ± 0.05	0.85 ± 0.03	0.59 ± 0.05
	75 mg	1.40 ± 0.15	1.17 ± 0.16	0.99 ± 0.12
	100 mg	0.74 ± 0.03	0.95 ± 0.05	0.48 ± 0.05
XI	50 mg	$0.22^* \pm 0.06$	$0.33^* \pm 0.14$	$0.33^* \pm 0.71$
	75 mg	$0.36^* \pm 0.11$	1.08 ± 0.11	0.53 ± 0.15
	100 mg	$0.11^* \pm 0.09$	0.94 ± 0.07	0.51 ± 0.18
XIII	50 mg	0.74 ± 0.04	0.62 ± 0.05	0.75 ± 0.03
	75 mg	1.04 ± 0.05	1.33 ± 0.06	0.90 ± 0.04
	100 mg	0.86 ± 0.05	1.03 ± 0.06	0.55 ± 0.03
Morphine HCl	10 mg	41.9 ± 0.52	44.8 ± 0.25	48.3 ± 0.30

N = 5 P < 0.05 (All the observations are significant except *Insignificant observations)

Table 3
Analgesic Effects of 7-Azaindole Derivatives (II, VII and XIV)
in Mouse Tail flick Method

Treatment	Dose/kg	Analgesia TFLD (mean increase in Latency after drug admn. $\pm$ SEM(s))		
		30 min	60 min	90 min
Water for Injection	0.3 ml	0.27 ± 0.16	0.31 ± 0.14	0.24 ± 0.19
II	50 mg	0.72 ± 0.08	0.69 ± 0.06	0.41 ± 0.04
	75 mg	0.93 ± 0.07	0.86 ± 0.05	0.85 ± 0.03
	100 mg	0.44 ± 0.03	0.79 ± 0.11	0.51 ± 0.02
VII	50 mg	$0.22^* \pm 0.11$	0.94 ± 0.05	1.06 ± 0.08
	75 mg	0.84 ± 0.06	0.88 ± 0.10	0.98 ± 0.13
	100 mg	$0.12^* \pm 0.08$	1.08 ± 0.13	1.09 ± 0.05
XIV	50 mg	$0.18^* \pm 0.01$	0.86 ± 0.08	0.69 ± 0.08
	75 mg	$0.22^* \pm 0.08$	1.02 ± 0.13	1.14 ± 0.06
	100 mg	$0.04^* \pm 0.11$	1.04 ± 0.08	0.86 ± 0.07
Morphine HCl	10 mg	43.1 ± 0.38	45.8 ± 0.69	49.3 ± 0.06

N = 5 P < 0.05 (All the observations are significant except *Insignificant observations)

Table 4
Analgesic Effects of 7-Azaindole Derivatives (IV-VI and XII)
in Mouse Tail Flick Method

Treatment	Dose/kg	Analgesia TFLD (mean increase in Latency after drug admn. $\pm$ SEM(s))		
		30 min	60 min	90 min
Water for Injection	0.3 ml	0.19 ± 0.05	0.17 ± 0.08	0.23 ± 0.08
IV	50 mg	0.48 ± 0.05	0.85 ± 0.04	0.38 ± 0.07
	75 mg	0.98 ± 0.06	1.21 ± 0.14	1.13 ± 0.06
	100 mg	0.56 ± 0.08	0.41 ± 0.04	0.49 ± 0.06
V	50 mg	0.61 ± 0.05	0.09 ± 0.05	0.91 ± 0.04
	75 mg	1.48 ± 0.04	2.06 ± 0.09	1.97 ± 0.08
	100 mg	1.41 ± 0.05	2.16 ± 0.06	1.85 ± 0.08
VI	50 mg	0.43 ± 0.04	1.24 ± 0.11	0.98 ± 0.11
	75 mg	1.43 ± 0.07	1.62 ± 0.06	1.36 ± 0.22
	100 mg	1.36 ± 0.04	2.01 ± 0.12	2.06 ± 0.17
XII	50 mg	$0.26^* \pm 0.1$	0.74 ± 0.06	0.56 ± 0.07
	75 mg	0.64 ± 0.06	0.92 ± 0.02	0.92 ± 0.09
	100 mg	$0.28^* \pm 0.16$	1.06 ± 0.08	1.14 ± 0.06
Morphine HCl	10 mg	46.2 ± 0.62	48.3 ± 0.42	52.9 ± 0.64

N = 5 P < 0.05 (All the observations are significant except *Insignificant observations)

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